

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0618

South Sheboygan Falls 138 kV
June 18, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0618. The Interconnection Customer proposes a 98.2 MW solar photovoltaic facility interconnecting at the South Sheboygan Falls 138 kV substation in the Kentucky area. The request was received on May 16, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 200 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,761,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WIXOM - ALBION RD 138 kV	DAVISVILLE - NORTH PPLETON 138 kV line	197.8	371	9.1
LESLIE - ST JOSEPH 138 kV line	ALBION ROAD - WIXOM 138 kV	129.6	444	11.7
LESLIE - NORTH APPLETON 138 kV Line	WIXOM - ALBION ROAD 138 kV	129.0	451	11.7
MIRABEL - DAVISVILLE 138 kV line	WELLINGTON - WEC 138 kV	128.9	308	5.1
NORTH APPLETON - DAVISVILLE 138 kV	ALBION RD - WIXOM 138 kV	128.2	460	11.7
MIRABEL - WEC 138 kV line	WELLINGTON SUBSTATION - WEC 138 kV	127.7	317	5.1
WELLINGTON - DAVISVILLE 138 kV	WIXOM - ALBION RD 138 kV	125.9	333	9.2
SAND SPRINGS - SNOHOMISH 138 kV line	SAND SPRINGS - OJO 138 kV Line	125.5	262	8.1
HAZARD - HARRIS 138 kV	MASSAPEQUA - WANSLEY COMBINED CYCLE 138 kV	125.0	227	5.3
CPU SILICON - LANHAM 138 kV	LIVERMORE - LEE COUNTY SOLID WASTE ENERGY 345 kV	124.8	175	14.4
SILVER BAY POWER - HOTEL RD 138 kV line	LIVERMORE - LEE COUNTY SOLID WASTE ENERGY 345 kV Line	124.4	179	12.2
SNOHOMISH - MAINWARING 138 kV	SAND SPRINGS - OJO 138 kV	123.7	259	8.1
HOLDEN SUBSTATION - EMBUDO 138 kV	LIVERMORE - LEE COUNTY SOLID WASTE ENERGY 345 kV line	121.9	236	14.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SPRING CRK RD - HARRIS 138 kV Line	MASSAPEQUA - WANSLEY COMBINED CYCLE 138 kV L	120.0	315	8.3
GREENVILLE AVE - BEANBLOSSOM-SW STA SUBSTATION 138 kV	LEE COUNTY SOLID WASTE ENERGY - LIVERMORE 345 kV	116.9	284	19.3
KINGS ROW - CPU SILICON 138 kV	LEE COUNTY SOLID WASTE ENERGY - LIVERMORE 345 kV	116.1	202	14.4
ANSON NORTH - CHEOAH 138 kV Line	WIXOM - ALBION RD 138 kV Line	115.9	247	5.4
SAND SPRINGS - OJO 138 kV	SAND SPRINGS - SNOHOMISH 138 kV	113.9	304	7.8
TEXACO BT - COLLINS 138 kV line	WIXOM - ALBION ROAD 138 kV	113.8	173	5.4
CHEOAH - SUMMIT AVE 138 kV Line	WIXOM - ALBION RD 138 kV	113.6	235	5.4
AMITY - CONSOLIDATED COAL 138 kV	SPRING CRK RD - HARRIS 138 kV	112.5	278	5.7
SUMMIT AVE - SONOITA 138 kV line	WIXOM - ALBION RD 138 kV line	112.4	231	5.4
DEL NORTE - SONOITA 138 kV	WIXOM - ALBION RD 138 kV L	111.4	228	5.4
AMITY - MASSAPEQUA 138 kV	HARRIS - SPRING CRK RD 138 kV Line	109.7	275	5.7
TEXACO BT - LETTERKENNY SUBSTATION 138 kV line	WIXOM - ALBION RD 138 kV Line	109.4	195	5.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WIXOM - ALBION RD 138 kV	Rebuild line with higher-capacity conductor	\$3,526,000	1% (\$49,000)
LESLIE SUBSTATION - ST JOSEPH 138 kV	Rebuild line with higher-capacity conductor	\$10,195,000	2% (\$214,000)
LESLIE - NORTH APPLETON 138 kV Line	Rebuild line with higher-capacity conductor	\$4,184,000	2% (\$88,000)
DAVISVILLE - MIRABEL SUBSTATION 138 kV line	Rebuild line with higher-capacity conductor	\$22,196,000	2% (\$358,000)
NORTH APPLETON - DAVISVILLE 138 kV	Rebuild line with higher-capacity conductor	\$4,361,000	2% (\$92,000)
MIRABEL - WEC 138 kV line	Rebuild line with higher-capacity conductor	\$6,558,000	2% (\$106,000)
WELLINGTON - DAVISVILLE 138 kV	Rebuild line with higher-capacity conductor	\$4,282,000	2% (\$96,000)
SAND SPRINGS - SNOHOMISH 138 kV	Rebuild line with higher-capacity conductor	\$26,537,000	3% (\$807,000)
HAZARD - HARRIS 138 kV	Rebuild line with higher-capacity conductor	\$29,486,000	3% (\$927,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CPU SILICON - LANHAM 138 kV L	Rebuild line with higher-capacity conductor	\$3,785,000	7% (\$272,000)
SILVER BAY POWER - HOTEL RD 138 kV line	Rebuild line with higher-capacity conductor	\$20,286,000	6% (\$1,245,000)
SNOHOMISH - MAINWARING 138 kV	Rebuild line with higher-capacity conductor	\$4,249,000	3% (\$129,000)
HOLDEN - EMBUDO 138 kV Line	Rebuild line with higher-capacity conductor	\$5,438,000	7% (\$391,000)
SPRING CRK RD - HARRIS 138 kV Line	Rebuild line with higher-capacity conductor	\$3,808,000	3% (\$107,000)
GREENVILLE AVE - BEANBLOSSOM-SW STA 138 kV	Rebuild line with higher-capacity conductor	\$15,762,000	8% (\$1,238,000)
KINGS ROW - CPU SILICON 138 kV	Rebuild line with higher-capacity conductor	\$13,116,000	7% (\$943,000)
ANSON NORTH - CHEOAH 138 kV Line	Rebuild line with higher-capacity conductor	\$2,681,000	3% (\$75,000)
SAND SPRINGS - OJO 138 kV	Rebuild line with higher-capacity conductor	\$10,820,000	3% (\$287,000)
COLLINS - TEXACO BT 138 kV	Rebuild line with higher-capacity conductor	\$3,784,000	3% (\$105,000)
CHEOAH - SUMMIT AVE 138 kV Line	Rebuild line with higher-capacity conductor	\$3,640,000	3% (\$101,000)
AMITY - CONSOLIDATED COAL 138 kV	Rebuild line with higher-capacity conductor	\$19,039,000	3% (\$545,000)
SONOITA - SUMMITAVE 138 kV	Rebuild line with higher-capacity conductor	\$6,426,000	3% (\$179,000)
DEL NORTE - SONOITA 138 kV line	Rebuild line with higher-capacity conductor	\$5,950,000	3% (\$165,000)
AMITY - MASSAPEQUA 138 kV	Reconductor line	\$1,830,000	3% (\$52,000)
LETTERKENNY - TEXACO BT 138 kV line	Reconductor line	\$6,826,000	3% (\$190,000)
	Total allocated to project		\$8,761,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.